Managing Vagueness in Semantic Web Languages

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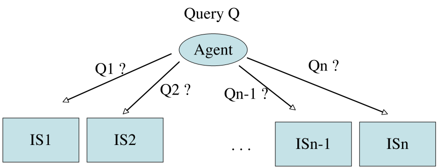
Sources of Uncertainty and Vagueness on the Web

- Information Retrieval:
  - To which degree is a Web site, a Web page, a text passage, an image region, a video segment, . . . relevant to my information need?
- Matchmaking
  - To which degree does an object match my requirements?
    - if I'm looking for a car and my budget is *about* 20.000 €, to which degree does a car's price of 20.500 € match my budget?

- Semantic annotation
    - To which degree does e.g., an image object represent a dog?
- Information extraction
    - To which degree am I'm sure that e.g., SW is an acronym of "Semantic Web"?
- Ontology alignment (schema mapping)
    - To which degree do two concepts of two ontologies represent the same, or are disjoint, or are overlapping?
- Representation of background knowledge
    - To some degree birds fly.
    - To some degree Jim is a blond and young.

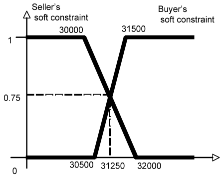
Example (Distributed Information Retrieval) [7]



Then the agent has to perform automatically the following steps:

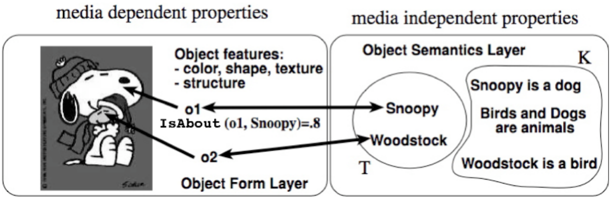
1. The agent has to select a subset of relevant resources $\mathscr{S}' \subseteq \mathscr{S}$, as it is not reasonable to assume to access to and query all resources (resource selection/resource discovery);
2. For every selected source $\mathcal{S}_i \in \mathscr{S}'$ the agent has to reformulate its information need Q_A into the query language $\mathcal{L}_i$ provided by the resource (schema mapping/ontology alignment);
3. The results from the selected resources have to be merged together (data fusion/rank aggregation)

Example (Negotiation) [2]



- A car seller sells an Audi TT for 31500 €, as from the catalog price.
- A buyer is looking for a sports-car, but wants to to pay not more than around 30000 €
- Classical DLs: the problem relies on the crisp conditions on price.
- More fine grained approach: to consider prices as vague constraints (fuzzy sets) (as usual in negotiation)
  - Seller would sell above 31500 €, but can go down to 30500 €
  - The buyer prefers to spend less than 30000 €, but can go up to 32000 €
  - Highest degree of matching is 0.75 . The car may be sold at 31250 €.

Example (Logic-based information retrieval model)[1, 8]



IsAbout		
ImageRegion	*Object ID*	*degree*
*o*1	*snoopy*	0.8
*o*2	*woodstock*	0.7
⋮	⋮	

"Find top-k image regions about animals"

$$Query(x) \leftarrow ImageRegion(x) \land isAbout(x, y) \land Animal(y)$$

Example (Database query) [3, 4, 5, 6]

HotelID	*hasLoc*
*h*1	*hl*1
*h*2	*hl*2
⋮	⋮

ConferenceID	*hasLoc*
*c*1	*cl*1
*c*2	*cl*2
⋮	⋮

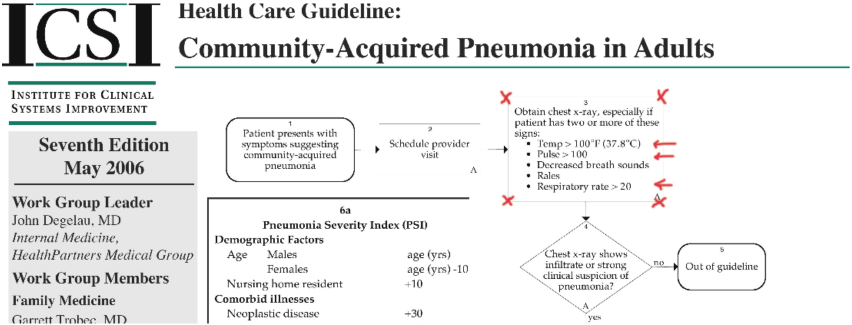
hasLoc	*hasLoc*	*distance*
*hl*1	*cl*1	300
*hl*1	*cl*2	500
*hl*2	*cl*1	750
*hl*2	*cl*2	800
⋮	⋮	

hasLoc	*hasLoc*	*close*	*cheap*
*hl*1	*cl*1	0.7	0.3
*hl*1	*cl*2	0.5	0.5
*hl*2	*cl*1	0.25	0.8
*hl*2	*cl*2	0.2	0.9
⋮	⋮	⋮	

"Find top-*k* cheapest hotels close to the train station"

$$q(h) \leftarrow hasLocation(h, hl) \land hasLocation(\mathtt{train}, cl) \land close(hl, cl) \land cheap(h)$$

Example (Health-care: diagnosis of pneumonia)



- E.g., $Temp = 37.5$, $Pulse = 98$, $RespiratoryRate = 18$ are in the “danger zone” already
- Temperature, Pulse and Respiratory rate, . . . : these constraints are rather imprecise than crisp

Uncertainty vs. Vagueness: a clarification

- What does the degree mean?
- There is often a misunderstanding between interpreting a degree as a measure of uncertainty or as a measure of vagueness
- The value 0.83 has a different interpretation in “Birds fly to degree 0.83” from that in “Hotel Verdi is close to the train station to degree 0.83”

Uncertainty

- Uncertainty: statements are true or false. But, due to lack of knowledge we can only estimate to which probability/possibility/necessity degree they are true or false
  - For instance, a bird flies or does not fly. The probability/possibility/necessity degree that it flies is 0.83
- Usually we have a possible world semantics with a distribution over possible worlds:

$$W = \{I \text{ classical interpretation}\}, \quad I(\varphi) \in \{0,1\}$$
$$\mu\colon W \to [0,1], \quad \mu(I) \in [0,1]$$
$$Pr(\phi) = \sum_{I \models \phi} \mu(I)$$
$$Poss(\phi) = \sup_{I \models \phi} \mu(I)$$
$$Necc(\phi) = \inf_{I \not\models \phi} \mu(I) = 1 - Poss(\neg\phi)$$

Vagueness

- Vagueness: statements involve concepts for which there is no exact definition, such as tall, small, close, far, cheap, expensive, isAbout, similarTo. Statements are true to some degree which is taken from a truth space.
    - E.g., "Hotel Verdi is close to the train station to degree 0.83"
- Truth space: set of truth values L and an partial order $\leq$
- Many-valued Interpretation: a function I mapping formulae into L, i.e. $I(\varphi) \in L$
- Fuzzy Logic: $L = [0, 1]$
- Uncertainty and Vagueness: "It is possible/probable to degree 0.83 that it will be hot tomorrow"
- The notion of imperfect information covers concepts such as uncertainty, vagueness, contradiction, incompleteness, imprecision.

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Web Ontology Languages

- Wide variety of languages for “Explicit Specification”
  - Graphical notations
    - Semantic networks
    - UML
    - RDF/RDFS
  - Logic based
    - Description Logics (e.g., OIL, DAML+OIL, OWL, OWL-DL, OWL-Lite)
    - Rules (e.g., RuleML, RIF, SWRL, LP/Prolog)
    - First Order Logic (e.g., KIF)
- RDF and OWL-DL are the major players (so far ...)

OWL [10]

- Three species of OWL
  - OWL full is union of OWL syntax and RDF (Undecidable)
  - OWL DL restricted to FOL fragment (decidable in NEXPTIME)
  - OWL Lite is "easier to implement" subset of OWL DL (decidable in EXPTIME)
- Semantic layering
  - OWL DL within Description Logic (DL) fragment
- OWL DL based on $\mathcal{SHOIN}(D_n)$ DL
- OWL Lite based on $\mathcal{SHIF}(D_n)$ DL

Description Logics (DLs)

- The logics behind OWL-DL and OWL-Lite, `http://dl.kr.org/`.
- Concept/Class: names are equivalent to unary predicates
  - In general, concepts equiv to formulae with one free variable
- Role or attribute: names are equivalent to binary predicates
  - In general, roles equiv to formulae with two free variables
- Taxonomy: Concept and role hierarchies can be expressed
- Individual: names are equivalent to constants
- Operators: used to build concepts/classes

The DL Family

- A given DL is defined by set of concept and role forming operators
- Basic language: $\mathcal{ALC}$(Attributive Language with Complement)

Syntax			Semantics	Example
C, D	$\rightarrow$	$\top$ |	$\top(x)$	
		$\bot$ |	$\bot(x)$	
		A |	$A(x)$	$Human$
		$C \sqcap D$ |	$C(x) \wedge D(x)$	$Human \sqcap Male$
		$C \sqcup D$ |	$C(x) \vee D(x)$	$Nice \sqcup Rich$
		$\neg C$ |	$\neg C(x)$	$\neg Meat$
		$\exists R.C$ |	$\exists y.R(x, y) \wedge C(y)$	$\exists has\_child.Blond$
		$\forall R.C$	$\forall y.R(x, y) \Rightarrow C(y)$	$\forall has\_child.Human$
$C \sqsubseteq D$			$\forall x.C(x) \Rightarrow D(x)$	$Happy\_Father \sqsubseteq Man \sqcap \exists has\_child.Female$
$a{:}C$			$C(a)$	$John{:}Happy\_Father$

Toy Example

$$
\begin{array}{rcl}
Sex & = & Male \sqcup Female \\
Male \sqcap Female & \sqsubseteq & \bot \\
Person & \sqsubseteq & Human \sqcap \exists hasSex.Sex \\
MalePerson & \sqsubseteq & Person \sqcap \exists hasSex.Male
\end{array}
$$

$$umberto{:}Person \sqcap \exists hasSex.\neg Female$$

$$KB \models umberto{:}MalePerson$$

Note on DL Naming

$\mathcal{AL}$: $C, D \longrightarrow \top \mid \bot \mid A \mid C \sqcap D \mid \neg A \mid \exists R.\top \mid \forall R.C$

$\mathcal{C}$: Concept negation, $\neg C$. Thus, $\mathcal{ALC} = \mathcal{AL} + \mathcal{C}$

$\mathcal{S}$: Used for $\mathcal{ALC}$ with transitive roles $\mathcal{R}_+$

$\mathcal{U}$: Concept disjunction, $C_1 \sqcup C_2$

$\mathcal{E}$: Existential quantification, $\exists R.C$

$\mathcal{H}$: Role inclusion axioms, $R_1 \sqsubseteq R_2$, e.g. $is\_component\_of \sqsubseteq is\_part\_of$

$\mathcal{N}$: Number restrictions, $(\geq n\ R)$ and $(\leq n\ R)$, e.g. $(\geq 3\ has\_Child)$ (has at least 3 children)

$\mathcal{Q}$: Qualified number restrictions, $(\geq n\ R.C)$ and $(\leq n\ R.C)$, e.g. $(\leq 2\ has\_Child.Adult)$ (has at most 2 adult children)

$\mathcal{O}$: Nominals (singleton class), $\{a\}$, e.g. $\exists has\_child.\{mary\}$.
Note: $a{:}C$ equiv to $\{a\} \sqsubseteq C$ and $(a, b){:}R$ equiv to $\{a\} \sqsubseteq \exists R.\{b\}$

$\mathcal{I}$: Inverse role, R^-, e.g. $isPartOf = hasPart^-$

$\mathcal{F}$: Functional role, f, e.g. $functional(hasAge)$

$\mathcal{R}_+$: transitive role, e.g. $transitive(isPartOf)$

For instance,

$$\mathcal{SHIF} = \mathcal{S} + \mathcal{H} + \mathcal{I} + \mathcal{F} = \mathcal{ALCR_+HIF} \quad \text{OWL-Lite (EXPTIME)}$$
$$\mathcal{SHOIN} = \mathcal{S} + \mathcal{H} + \mathcal{O} + \mathcal{I} + \mathcal{N} = \mathcal{ALCR_+HOIN} \quad \text{OWL-DL (NEXPTIME)}$$

Semantics of Additional Constructs

$\mathcal{H}$: Role inclusion axioms, $\mathcal{I} \models R_1 \sqsubseteq R_2$ iff ${R_1}^{\mathcal{I}} \subseteq {R_1}^{\mathcal{I}}$

$\mathcal{N}$: Number restrictions,
$(\geq n\ R)^{\mathcal{I}} = \{x \in \Delta^{\mathcal{I}} : |\{y \mid \langle x, y \rangle \in R^{\mathcal{I}}\}| \geq n\}$,
$(\leq n\ R)^{\mathcal{I}} = \{x \in \Delta^{\mathcal{I}} : |\{y \mid \langle x, y \rangle \in R^{\mathcal{I}}\}| \leq n\}$

$\mathcal{Q}$: Qualified number restrictions,
$(\geq n\ R.C)^{\mathcal{I}} = \{x \in |\{y \mid \langle x, y \rangle \in R^{\mathcal{I}} \wedge y \in C^{\mathcal{I}}\}| \geq n\}$,
$(\leq n\ R.C)^{\mathcal{I}} = \{x \in \Delta^{\mathcal{I}} : |\{y \mid \langle x, y \rangle \in R^{\mathcal{I}} \wedge y \in C^{\mathcal{I}}\}| \leq n\}$

$\mathcal{O}$: Nominals (singleton class), $\{a\}^{\mathcal{I}} = \{a^{\mathcal{I}}\}$

$\mathcal{I}$: Inverse role, $(R^{-})^{\mathcal{I}} = \{\langle x, y \rangle \mid \langle y, x \rangle \in R^{\mathcal{I}}\}$

$\mathcal{F}$: Functional role, $I \models fun(f)$ iff $\forall z \forall y \forall z$ if $\langle x, y \rangle \in f^{\mathcal{I}}$ and $\langle x, z \rangle \in f^{\mathcal{I}}$ the $y = z$

$\mathcal{R}_{+}$: transitive role,
$(R_{+})^{\mathcal{I}} = \{\langle x, y \rangle \mid \exists z$ such that $\langle x, z \rangle \in R^{\mathcal{I}} \wedge \langle z, y \rangle \in R^{\mathcal{I}}\}$

Concrete Domains

- Concrete domains: reals, integers, strings, ...

$$(tim, 14){:}hasAge$$
$$(sf, \text{``}SoftComputing\text{''}){:}hasAcronym$$
$$(source1, \text{``}ComputerScience\text{''}){:}isAbout$$
$$(service2, \text{``}InformationRetrievalTool''){:}Matches$$
$$Minor = Person \sqcap \exists hasAge. \leq_{18}$$

- Semantics: a clean separation between "object" classes and concrete domains
  - $D = \langle \Delta_D, \Phi_D \rangle$
  - Δ_D is an interpretation domain
  - Φ_D is the set of concrete domain predicates d with a predefined arity n and fixed interpretation $d^D \subseteq \Delta_D^n$
  - Concrete properties: $R^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \Delta_D$
- Notation: (D). E.g., $\mathcal{ALC}(D)$ is $\mathcal{ALC}$ + concrete domains

OWL DL as Description Logic

Concept/Class constructors:

Abstract Syntax	DL Syntax	Example
Descriptions (C)		
A (URI reference)	A	`Conference`
`owl:Thing`	$\top$	
`owl:Nothing`	$\bot$	
`intersectionOf`(C_1 C_2 ...)	$C_1 \sqcap C_2$	`Reference` $\sqcap$ `Journal`
`unionOf`(C_1 C_2 ...)	$C_1 \sqcup C_2$	`Organization` $\sqcup$ `Institution`
`complementOf`(C)	$\neg C$	$\neg$ `MasterThesis`
`oneOf`(o_1 ...)	$\{o_1, \ldots\}$	`{"WISE","ISWC",...}`
`restriction`(R `someValuesFrom`(C))	$\exists R.C$	$\exists$`parts.InCollection`
`restriction`(R `allValuesFrom`(C))	$\forall R.C$	$\forall$`date.Date`
`restriction`(R `hasValue`(o))	$\exists R.\{o\}$	$\exists$`date.{2005}`
`restriction`(R `minCardinality`(n))	$(\geq n\ R)$	($\geqslant$ 1 `location`)
`restriction`(R `maxCardinality`(n))	$(\leq n\ R)$	($\leqslant$ 1 `publisher`)
`restriction`(U `someValuesFrom`(D))	$\exists U.D$	$\exists$`issue.integer`
`restriction`(U `allValuesFrom`(D))	$\forall U.D$	$\forall$`name.string`
`restriction`(U `hasValue`(v))	$\exists U.=_v\}$	$\exists$`series.`$=_{\text{"LNCS"}}$
`restriction`(U `minCardinality`(n))	$(\geq n\ U)$	($\geqslant$ 1 `title`)
`restriction`(U `maxCardinality`(n))	$(\leq n\ U)$	($\leqslant$ 1 `author`)

Note: R is an abstract role, while U is a concrete property of arity two.

Axioms:

Abstract Syntax	DL Syntax	Example
Axioms		
`Class`(A `partial` $C_1 \ldots C_n$)	$A \sqsubseteq C_1 \sqcap \ldots \sqcap C_n$	$Human \sqsubseteq Animal \sqcap Biped$
`Class`(A `complete` $C_1 \ldots C_n$)	$A = C_1 \sqcap \ldots \sqcap C_n$	$Man = Human \sqcap Male$
`EnumeratedClass`($A\ o_1 \ldots o_n$)	$A = \{o_1\} \sqcup \ldots \sqcup \{o_n\}$	$RGB = \{r\} \sqcup \{g\} \sqcup \{b\}$
`SubClassOf`($C_1\ C_2$)	$C_1 \sqsubseteq C_2$	
`EquivalentClasses`($C_1 \ldots C_n$)	$C_1 = \ldots = C_n$	
`DisjointClasses`($C_1 \ldots C_n$)	$C_i \sqcap C_j = \perp, i \neq j$	$Male \sqcap Female \sqsubseteq \perp$
`ObjectProperty`(R `super` (R_1)... `super` (R_n)	$R \sqsubseteq R_i$	$HasDaughter \sqsubseteq hasChild$
`domain`(C_1) ...`domain`(C_n)	$(\geq 1\ R) \sqsubseteq C_i$	$(\geq 1\ hasChild) \sqsubseteq Human$
`range`(C_1) ...`range`(C_n)	$\top \sqsubseteq \forall R.C_i$	$\top \sqsubseteq \forall hasChild.Human$
[`inverseof`(P)]	$R = P^-$	$hasChild = hasParent^-$
[`symmetric`]	$R = R^-$	$similar = similar^-$
[`functional`]	$\top \sqsubseteq (\leq 1\ R)$	$\top \sqsubseteq (\leq 1\ hasMother)$
[`Inversefunctional`]	$\top \sqsubseteq (\leq 1\ R^-)$	
[`Transitive`])	$Tr(R)$	$Tr(ancestor)$
`SubPropertyOf`($R_1\ R_2$)	$R_1 \sqsubseteq R_2$	
`EquivalentProperties`($R_1 \ldots R_n$)	$R_1 = \ldots = R_n$	$cost = price$
`AnnotationProperty`(S)		

Abstract Syntax	DL Syntax	Example
DatatypeProperty(U super (U_1)... super (U_n)	$U \sqsubseteq U_i$	
domain(C_1)...domain(C_n)	$(\geq 1\ U) \sqsubseteq C_i$	$(\geq 1\ hasAge) \sqsubseteq Human$
range(D_1)...range(D_n)	$\top \sqsubseteq \forall U.D_i$	$\top \sqsubseteq \forall hasAge.posInteger$
[functional])	$\top \sqsubseteq (\leq 1\ U)$	$\top \sqsubseteq (\leq 1\ hasAge)$
SubPropertyOf($U_1 U_2$)	$U_1 \sqsubseteq U_2$	$hasName \sqsubseteq hasFirstName$
EquivalentProperties($U_1 \ldots U_n$)	$U_1 = \ldots = U_n$	
Individuals		
Individual(o type (C_1)... type (C_n))	o:C_i	tim:$Human$
value($R_1 o_1$)...value($R_n o_n$)	(o, o_i):R_i	$(tim, mary)$:$hasChild$
value($U_1 v_1$)...value($U_n v_n$)	(o, v_1):U_i	$(tim, 14)$:$hasAge$
SameIndividual($o_1 \ldots o_n$)	$o_1 = \ldots = o_n$	$president\_Bush = G.W.Bush$
DifferentIndividuals($o_1 \ldots o_n$)	$o_i \neq o_j, i \neq j$	$john \neq peter$
Symbols		
Object Property R (URI reference)	R	hasChild
Datatype Property U (URI reference)	U	hasAge
Individual o (URI reference)	U	tim
Data Value v (RDF literal)	U	“ESWC07”

LPs Basics (for ease, without default negation) [6]

- Predicates are *n*-ary
- Terms are variables or constants
- Rules are of the form

$$P(\mathbf{x}) \leftarrow \varphi(\mathbf{x}, \mathbf{y})$$

  where $\varphi(\mathbf{x}, \mathbf{y})$ is a formula built from atoms of the form $B(\mathbf{z})$ and connectors $\wedge, \vee$
  For instance,

$$has\_father(x, y) \quad \leftarrow \quad has\_parent(x, y) \wedge Male(y)$$

- Facts are rules with empty body
  For instance,

$$has\_parent(mary, jo)$$

LPs Semantics: FOL semantics

- $\mathcal{P}^*$ is constructed as follows:
  1. set $\mathcal{P}^*$ to the set of all ground instantiations of rules in $\mathcal{P}$;
  2. if atom A is not head of any rule in $\mathcal{P}^*$, then add $A \leftarrow 0$ to $\mathcal{P}^*$;
  3. replace several rules in $\mathcal{P}^*$ having same head

$$\left.\begin{array}{l} A \leftarrow \varphi_1 \\ A \leftarrow \varphi_2 \\ \vdots \\ A \leftarrow \varphi_n \end{array}\right\} \text{ with } A \leftarrow \varphi_1 \vee \varphi_2 \vee \ldots \vee \varphi_n \; .$$

- Note: in $\mathcal{P}^*$ each atom $A \in B_\mathcal{P}$ is head of exactly one rule
- Herbrand Base of $\mathcal{P}$ is the set $B_\mathcal{P}$ of ground atoms
- Interpretation is a function $I : B_\mathcal{P} \to \{0, 1\}$.
- Model $I \models \mathcal{P}$ iff for all $r \in \mathcal{P}^*$ $I \models r$, where $I \models A \leftarrow \varphi$ iff $I(\varphi) \leq I(A)$
- Least model exists and is least fixed-point of

$$T_\mathcal{P}(I)(A) = I(\varphi), \text{ for all } A \leftarrow \varphi \in \mathcal{P}^*$$

Toy Example

$$\begin{array}{rcl} Q(x) & \leftarrow & B(x) \\ Q(x) & \leftarrow & C(x) \\ B(a) & \leftarrow & \\ C(b) & \leftarrow & \end{array}$$

$$KB \models Q(a) \quad KB \models Q(b) \qquad answers(KB, Q) = \{a, b\}$$

where $answers(KB, Q) = \{\mathbf{c} \mid KB \models Q(\mathbf{c})\}$

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Vagueness

- Vagueness: statements involve concepts for which there is no exact definition, such as tall, close, cheap, IsAbout, simialarTo ...
- Statements are true to some degree which is taken from a truth space
  - E.g., "Hotel Verdi is close to the train station to degree 0.83"
  - "Find top-k cheapest hotels close to the train station"

$$q(h) \leftarrow hasLocation(h, hl) \wedge hasLocation(\mathtt{train}, cl) \wedge close(hl, cl) \wedge cheap(h)$$

- Truth space: usually $[0, 1]$
- Interpretation: a function I mapping atoms into $[0, 1]$, i.e. $I(A) \in [0, 1]$
- Problem: what is the interpretation of e.g. $close(verdi, train) \wedge cheap(200)$?
  - E.g., if $I(close(verdi, train)) = 0.83$ and $I(cheap(200)) = 0.2$, what is the result of $0.83 \wedge 0.2$?
  - E.g., In multimedia retrieval: if a image region is white to degree 0.8 and the object is about a dog to degree 0.4, to which degree is the image about a "white dog"? That is, what is $0.8 \wedge 0.4$?
- More generally, what is the result of $n \wedge m$, for $n, m \in [0, 1]$?
- The choice cannot be any arbitrary computable function, but has to reflect some basic properties that one expects to hold for a "conjunction"

Propositional Fuzzy Logics Basics [5]

- Formulae: propositional formulae
- Truth space is $[0, 1]$
- Formulae have a a degree of truth in $[0, 1]$
- Interpretation: is a mapping $\mathcal{I} : \mathit{Atoms} \to [0, 1]$

- Interpretations are extended to formulae using norms to interpret connectives $\wedge, \vee, \neg, \rightarrow$

negation
$n(0) = 1$
$a \leq b$ implies $n(b) \leq n(a)$

t-norm (conjunction)
$t(a, 1) = a$
$b \leq c$ implies $t(a, b) \leq t(a, c)$
$t(a, b) = t(b, a)$
$t(a, t(b, c)) = t(t(a, b), c)$

s-norm (disjunction)
$s(a, 0) = a$
$b \leq c$ implies $s(a, b) \leq s(a, c)$
$s(a, b) = s(b, a)$
$s(a, s(b, c)) = s(s(a, b), c)$

i-norm (implication)
$a \leq b$ implies $i(a, c) \geq i(b, c)$
$b \leq c$ implies $i(a, b) \leq i(a, c)$
$i(0, b) = 1$
$i(a, 1) = 1$

Usually,
$i(a, b) = \sup\{c \colon t(a, c) \leq b\}$

$i(a, b) = \sup\{c \colon t(a, c) \leq b\}$ is called r-implication and depends on the t-norm only

Typical norms

	Lukasiewicz Logic	Gödel Logic	Product Logic	Zadeh
$\neg x$	$1 - x$	if $x = 0$ then 1 else 0	if $x = 0$ then 1 else 0	$1 - x$
$x \wedge y$	$\max(x + y - 1, 0)$	$\min(x, y)$	$x \cdot y$	$\min(x, y)$
$x \vee y$	$\min(x + y, 1)$	$\max(x, y)$	$x + y - x \cdot y$	$\max(x, y)$
$x \Rightarrow y$	if $x \leq y$ then 1 else $1 - x + y$	if $x \leq y$ then 1 else y	if $x \leq y$ then 1 else y/x	$\max(1 - x, y)$

Note: for Lukasiewicz Logic and Zadeh, $x \Rightarrow y \equiv \neg x \vee y$

$$\begin{aligned}
\mathcal{I}(\phi \wedge \psi) &= \mathcal{I}(\phi) \wedge \mathcal{I}(\psi) \\
\mathcal{I}(\phi \vee \psi) &= \mathcal{I}(\phi) \vee \mathcal{I}(\psi) \\
\mathcal{I}(\phi \rightarrow \psi) &= \mathcal{I}(\phi) \rightarrow \mathcal{I}(\psi)
\end{aligned}$$

$$\begin{array}{rcl}
\mathcal{I} \models \phi & \text{iff} & \mathcal{I}(\phi) = 1 \quad \text{iff } \phi \text{ satisfiable} \\
\mathcal{I} \models \mathcal{T} & \text{iff} & \mathcal{I} \models \phi \text{ for all } \phi \in \mathcal{T} \\
\models \phi & \text{iff} & \text{for all } \mathcal{I} . \mathcal{I} \models \phi \\
\mathcal{T} \models \phi & \text{iff} & \text{for all } \mathcal{I}. \text{ if } \mathcal{I} \models \mathcal{T} \text{ then } \mathcal{I} \models \phi
\end{array}$$

- Note:

$$
\begin{array}{rcl}
\neg\phi & \text{is} & \phi \rightarrow 0 \\
\phi \bar{\wedge} \psi & \text{defined as} & \phi \wedge (\phi \rightarrow \psi) \\
\phi \bar{\vee} \psi & \text{defined as} & ((\phi \rightarrow \psi) \rightarrow \psi) \bar{\wedge} ((\psi \rightarrow \phi) \rightarrow \phi) \\
\mathcal{I}(\phi \bar{\wedge} \psi) & = & \min(\mathcal{I}(\phi), \mathcal{I}(\psi)) \\
\mathcal{I}(\phi \bar{\vee} \psi) & = & \max(\mathcal{I}(\phi), \mathcal{I}(\psi))
\end{array}
$$

- Zadeh semantics: not interesting for fuzzy logicians: its a sub-logic of Łukasiewicz and, thus, rarely considered by fuzzy logicians

$$
\begin{array}{rcl}
\neg_Z \phi & = & \neg_Ł \phi \\
\phi \wedge_Z \psi & = & \phi \wedge_Ł (\phi \rightarrow_Ł \psi) \\
\phi \rightarrow_Z \psi & = & \neg_Ł \phi \vee_Ł \psi
\end{array}
$$

Some additional properties of t-norms, s-norms, implication functions, and negation functions of various fuzzy logics.

Łukasiewicz logic	Gödel logic	Product logic	Zadeh logic
$x \wedge \neg x = 0$	$\exists x.\ x \wedge \neg x \neq 0$	$\exists x.\ x \wedge \neg x \neq 0$	$\exists x.\ x \wedge \neg x \neq 0$
$x \vee \neg x = 1$	$\exists x.\ x \vee \neg x \neq 1$	$\exists x.\ x \vee \neg x \neq 1$	$\exists x.\ x \vee \neg x \neq 1$
$\exists x.\ x \wedge x \neq x$	$x \wedge x = x$	$\exists x.\ x \wedge x \neq x$	$x \wedge x = x$
$\exists x.\ x \vee x \neq x$	$x \vee x = x$	$\exists x.\ x \vee x \neq x$	$x \vee x = x$
$\neg\neg x = x$	$\exists x.\ \neg\neg x \neq x$	$\exists x.\ \neg\neg x \neq x$	$\neg\neg x = x$
$x \rightarrow y = \neg x \vee y$	$\exists x.\ x \rightarrow y \neq \neg x \vee y$	$\exists x.\ x \rightarrow y \neq \neg x \vee y$	$x \rightarrow y = \neg x \vee y$
$\neg(x \rightarrow y) = x \wedge \neg y$	$\exists x.\ \neg(x \rightarrow y) \neq x \wedge \neg y$	$\exists x.\ \neg(x \rightarrow y) \neq x \wedge \neg y$	$\neg(x \rightarrow y) = x \wedge \neg y$
$\neg(x \wedge y) = \neg x \vee \neg y$	$\neg(x \wedge y) = \neg x \vee \neg y$	$\neg(x \wedge y) = \neg x \vee \neg y$	$\neg(x \wedge y) = \neg x \vee \neg y$
$\neg(x \vee y) = \neg x \wedge \neg y$	$\neg(x \vee y) = \neg x \wedge \neg y$	$\neg(x \vee y) = \neg x \wedge \neg y$	$\neg(x \vee y) = \neg x \wedge \neg y$

Axioms of logic BL (Basic Fuzzy Logic)

Fix arbitray t-norm and r-implication.

(A1) $(\phi \to \psi) \to ((\psi \to \chi) \to \phi \to \chi)$
(A2) $(\phi \wedge \psi) \to \phi$
(A3) $(\phi \wedge \psi) \to (\psi \wedge \phi)$
(A4) $(\phi \wedge (\phi \to \psi)) \to (\psi \wedge (\psi \to \phi))$
(A5a) $(\phi \wedge (\psi \to \chi)) \to ((\phi \wedge \psi) \to \chi))$
(A5b) $((\phi \wedge \psi) \to \chi)) \to (\phi \wedge (\psi \to \chi))$
(A6) $(\phi \wedge (\psi \to \chi)) \to (((\psi \to \phi) \to \chi)) \to \chi)$
(A7) $0 \to \phi$
(Deduction rule) Modus ponens: from ϕ and $\phi \to \psi$ infer ψ

Proposition

$\mathcal{T} \vdash_{BL} \phi$ iff $\mathcal{T} \models_{BL} \phi$. Also, if $\mathcal{T} \vdash_{BL} \phi$ then $\mathcal{T} \models_{BL2} \phi$, but not vice-versa (e.g. $\models_{BL2} \phi \vee \neg\phi$, but $\not\models_{BL} \phi \vee \neg\phi$).

- $\models_{BL} \phi \wedge \neg\phi \to 0$
- $\models_{BL} \phi \to \neg\neg\phi$, but $\not\models_{BL} \neg\neg\phi \to \phi$, e.g. $\phi = p \vee \neg p$, t-norm is Gödel
- $\models_{BL} (\phi \to \psi) \to (\neg\psi \to \neg\phi)$, but not vice-versa

Axioms of Łukasiewicz logic Ł

Fix Łukasiewicz t-norm and r-implication.

(Axioms)	Axioms of BL
(Ł)	$\neg\neg\phi \to \phi$
(Deduction rule)	Modus ponens: from ϕ and $\phi \to \psi$ infer ψ

Proposition

$\mathcal{T} \vdash_{Ł} \phi$ *iff* $\mathcal{T} \models_{Ł} \phi$.

- $\models_{Ł} \phi \to \psi \equiv \neg\psi \to \neg\phi$
- $\models_{Ł} \neg(\phi \wedge \psi) \equiv \neg\phi \vee \neg\psi$
- $\models_{Ł} \phi \to \psi \equiv \neg(\phi \wedge \neg\psi)$
- $\models_{Ł} \phi \to \psi \equiv \neg\phi \vee \neg\psi$
- $\models_{Ł} \neg(\phi \to \psi) \equiv \phi \wedge \neg\psi$
- Recall that “Zadeh logic” is a sub-logic of Ł

Axioms of Product logic Π

Fix product t-norm and r-implication.

(Axioms) Axioms of BL

(Π1) $\neg\neg\chi \to ((\phi \wedge \chi \to \psi \wedge \chi) \to (\phi \to \psi))$

(Π2) $(\phi \bar{\wedge} \neg\phi) \to 0$

(Deduction rule) Modus ponens: from ϕ and $\phi \to \psi$ infer ψ

Proposition

$\mathcal{T} \vdash_{\Pi} \phi$ *iff* $\mathcal{T} \models_{\Pi} \phi$.

- $\models_{\Pi} \neg(\phi \wedge \psi) \to \neg(\phi \bar{\wedge} \psi)$
- $\models_{\Pi} (\phi \to \neg\phi) \to \neg\phi$
- $\models_{\Pi} \neg\phi \bar{\vee} \neg\neg\phi$

Axioms of Gödel logic G

Fix Gödel t-norm and r-implication.

(Axioms) Axioms of BL

(G) $\phi \to (\phi \wedge \phi)$

(Deduction rule) Modus ponens: from ϕ and $\phi \to \psi$ infer ψ

Proposition

$\mathcal{T} \vdash_G \phi$ *iff* $\mathcal{T} \models_G \phi$.

- $\models_G (\phi \wedge \psi) \equiv (\phi \bar{\wedge} \psi)$
- Gödel logic proves all axioms of intuitionistic logic I, vice-versa I + (A6) proves all axioms of Gödel logic

Axioms of Boolean logic

Fix interpretations to be boolean.

(Axioms) Axioms of BL

(BL2) $\phi \bar{\vee} \neg \phi$

(Deduction rule) Modus ponens: from ϕ and $\phi \to \psi$ infer ψ

Proposition

$\mathcal{T} \vdash_{BL2} \phi$ iff $\mathcal{T} \models_{BL2} \phi$.

- $\models_{BL2} \phi \to (\phi \wedge \phi)$ (BL2 extends G)
- Ł + G is equivalent to BL2
- Ł + Π is equivalent to BL2
- G + Π is equivalent to BL2

Axioms of Rational Pavelka Logic (RPL)

- Fix Łukasiewicz t-norm and r-implication
- Rational $r \in [0,1]$ may appear as atom in formula. $\mathcal{I}(r) = r$
- Note: $\mathcal{I}(r \rightarrow \phi) = 1$ iff $\mathcal{I}(\phi) \geq r$. Also, $\mathcal{I}(\phi \rightarrow r) = 1$ iff $\mathcal{I}(\phi) \leq r$

(Axioms) Axioms of Ł

(Deduction rule) Modus ponens: from ϕ and $\phi \rightarrow \psi$ infer ψ

Proposition

$\mathcal{T} \vdash_{RPL} \phi$ iff $\mathcal{T} \models_{RPL} \phi$.

- RPL proves the derived deduction rule ($r, s \in [0,1]$): from $r \rightarrow \phi$ and $s \rightarrow (\phi \rightarrow \psi)$ infer $(r \wedge s) \rightarrow \psi$

  From $\phi \geq r$ and $(\phi \rightarrow \psi) \geq s$ infer $\psi \geq r \wedge s$

- Let

$$\begin{array}{lcl} ||\phi||_{\mathcal{T}} & = & \inf\{\mathcal{I}(\phi) \mid \mathcal{I} \models \mathcal{T}\} \text{ (truth degree)} \\ |\phi|_{\mathcal{T}} & = & \sup\{r \mid \mathcal{T} \vdash r \rightarrow \phi\} \text{ (provability degree)} \end{array}$$

  then $||\phi||_{\mathcal{T}} = |\phi|_{\mathcal{T}}$

- Also,

$$\begin{array}{lcl} |\neg\phi|_{\mathcal{T}} & = & 1 - |\phi|_{\mathcal{T}}| \\ |\phi|_{\mathcal{T}}| = \sup\{r \mid \mathcal{T} \vdash r \rightarrow \phi\} & = & \inf\{s \mid \mathcal{T} \vdash \phi \rightarrow s\} \end{array}$$

Predicate Fuzzy Logics Basics [5]

- Formulae: First-Order Logic formulae, *terms* are either variables or constants
  - we may introduce functions symbols as well, with crisp semantics (but uninteresting), or we need to discuss also fuzzy equality (which we leave out here)
- Truth space is $[0, 1]$
- Formulae have a a degree of truth in $[0, 1]$
- Interpretation: is a mapping $\mathcal{I} : \mathit{Atoms} \to [0, 1]$
- Interpretations are extended to formulae as follows:

$$
\begin{array}{rcl}
\mathcal{I}(\neg\phi) & = & \mathcal{I}(\phi) \to 0 \\
\mathcal{I}(\phi \land \psi) & = & \mathcal{I}(\phi) \land \mathcal{I}(\psi) \\
\mathcal{I}(\phi \to \psi) & = & \mathcal{I}(\phi) \to \mathcal{I}(\psi) \\
\mathcal{I}(\exists x \phi) & = & \sup_{c \in \Delta^{\mathcal{I}}} \mathcal{I}^c_x(\phi) \\
\mathcal{I}(\forall x \phi) & = & \inf_{c \in \Delta^{\mathcal{I}}} \mathcal{I}^c_x(\phi)
\end{array}
$$

  where $\mathcal{I}^c_x$ is as $\mathcal{I}$, except that variable x is mapped into individual c
- Definitions of $\mathcal{I} \models \phi$, $\mathcal{I} \models \mathcal{T}$, $\models \phi$, $\mathcal{T} \models \phi$, $||\phi||_{\mathcal{T}}$ and $|\phi|_{\mathcal{T}}$ are as for the propositional case

Axioms of logic $\mathcal{C}\forall$, where $\mathcal{C} \in \{$BL,Ł,Π,G$\}$

(Axioms)	Axioms of $\mathcal{C}$
($\forall$1)	$\forall x \phi(x) \rightarrow \phi(t)$ (t substitutable for x in $\phi(x)$)
($\exists$1)	$\phi(t) \rightarrow \exists x \phi(x)$ (t substitutable for x in $\phi(x)$)
($\forall$2)	$\forall x(\psi \rightarrow \phi) \rightarrow (\psi \rightarrow \forall x \phi)$ (x not free in ψ)
($\exists$2)	$\forall x(\phi \rightarrow \psi) \rightarrow (\exists x \phi \rightarrow \psi)$ (x not free in ψ)
($\forall$3)	$\forall x(\phi \bar{\vee} \psi) \rightarrow (\forall x \phi) \bar{\vee} \psi$ (x not free in ψ)
(Modus ponens)	from ϕ and $\phi \rightarrow \psi$ infer ψ
(Generalization)	from ϕ infer $\forall x \phi$

Proposition

$\mathcal{T} \vdash_{\mathcal{C}} \phi$ *iff* $\mathcal{T} \models_{\mathcal{C}} \phi$.

- if $\rightarrow$ is an r-implication then $||\psi||_{\mathcal{T}} \geq ||\phi||_{\mathcal{T}} \wedge ||\phi \rightarrow \psi||_{\mathcal{T}}$
- $\models_{BL\forall} \exists x \phi \rightarrow \neg \forall x \neg \phi$
- $\models_{BL\forall} \neg \exists x \phi \equiv \forall x \neg \phi$
- $\models_{Ł\forall} \exists x \phi \equiv \neg \forall x \neg \phi$

- $(\neg\forall x p(x)) \wedge (\neg\exists x \neg p(x))$ has no classical model. In Gödel logic it has no finite model, but has an infinite model: for integer $n \geq 1$, let $\mathcal{I}$ such that $p^{\mathcal{I}}(n) = 1/n$

$$\begin{array}{rcl} (\forall x p(x))^{\mathcal{I}} & = & \inf_n 1/n = 0 \\ (\exists x \neg p(x))^{\mathcal{I}} & = & \sup_n \neg 1/n = \sup 0 = 0 \end{array}$$

- Note: If $\mathcal{I} \models \exists x \phi(x)$ then not necessarily there is $c \in \Delta^{\mathcal{I}}$ such that $\mathcal{I} \models \phi(c)$.

$$\begin{array}{rcl} \Delta^{\mathcal{I}} & = & \{n \mid \text{ integer } n \geq 1\} \\ p^{\mathcal{I}}(n) & = & 1 - 1/n < 1, \text{ for all } n \\ (\exists x p(x))^{\mathcal{I}} & = & \sup_n 1 - 1/n = 1 \end{array}$$

- Witnessed formula: $\exists x \phi(x)$ is witnessed in $\mathcal{I}$ iff there is $c \in \Delta^{\mathcal{I}}$ such that $(\exists x \phi(x))^{\mathcal{I}} = (\phi(c))^{\mathcal{I}}$ (similarly for $\forall x \phi(x)$)
- Witnessed interpretation: $\mathcal{I}$ witnessed if all quantified formulae are witnessed in $\mathcal{I}$

Proposition

In Ł, ϕ is satisfiable iff there is a witnessed model of ϕ.

The proposition does not hold for logic G and Π

Predicate Rational Pavelka Logic (RPL∀)

- Fix Łukasiewicz t-norm and r-implication
- Formulae are as for Ł∀, where rationals $r \in [0, 1]$ may appear as atoms

(Axioms and rules) As for Ł∀

Proposition

$\mathcal{T} \vdash_{RPL\forall} \phi$ iff $\mathcal{T} \models_{RPL\forall} \phi$.

Fuzzy DLs Basics [26]

The semantics is an immediate consequence of the First-Order-Logic translation of DLs expressions

Interpretation:

$$\mathcal{I} = \Delta^{\mathcal{I}} \qquad C^{\mathcal{I}} : \Delta^{\mathcal{I}} \to [0,1] \qquad R^{\mathcal{I}} : \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \to [0,1]$$

$\wedge$ = t-norm
$\vee$ = s-norm
$\neg$ = negation
$\to$ = implication

Concepts:

Syntax		Semantics		
$C, D \longrightarrow$	$\top$ |	$\top^{\mathcal{I}}(x)$	$=$	1
	$\bot$ |	$\bot^{\mathcal{I}}(x)$	$=$	0
	A |	$A^{\mathcal{I}}(x)$	$\in$	$[0,1]$
	$C \sqcap D$ |	$(C_1 \sqcap C_2)^{\mathcal{I}}(x)$	$=$	$C_1{}^{\mathcal{I}}(x) \wedge C_2{}^{\mathcal{I}}(x)$
	$C \sqcup D$ |	$(C_1 \sqcup C_2)^{\mathcal{I}}(x)$	$=$	$C_1{}^{\mathcal{I}}(x) \vee C_2{}^{\mathcal{I}}(x)$
	$\neg C$ |	$(\neg C)^{\mathcal{I}}(x)$	$=$	$\neg C^{\mathcal{I}}(x)$
	$\exists R.C$ |	$(\exists R.C)^{\mathcal{I}}(x)$	$=$	$\sup_{y \in \Delta^{\mathcal{I}}} R^{\mathcal{I}}(x,y) \wedge C^{\mathcal{I}}(y)$
	$\forall R.C$	$(\forall R.C)^{\mathcal{I}}(u)$	$=$	$\inf_{y \in \Delta^{\mathcal{I}}} R^{\mathcal{I}}(x,y) \to C^{\mathcal{I}}(y)\}$

Assertions: $\langle a{:}C, r\rangle$, $\mathcal{I} \models \langle a{:}C, r\rangle$ iff $C^{\mathcal{I}}(a^{\mathcal{I}}) \geq r$ (similarly for roles)

- individual a is instance of concept C at least to degree r, $r \in [0,1] \cap \mathbb{Q}$

Inclusion axioms: $C \sqsubseteq D$,

- $\mathcal{I} \models C \sqsubseteq D$ iff $\forall x \in \Delta^{\mathcal{I}}.C^{\mathcal{I}}(x) \leq D^{\mathcal{I}}(x)$
- this is equivalent to, $\forall x \in \Delta^{\mathcal{I}}.(C^{\mathcal{I}}(x) \to D^{\mathcal{I}}(x)) = 1$, if $\to$ is an r-implication

Basic Inference Problems

Consistency: Check if knowledge is meaningful

- Is *KB* consistent, i.e. satisfiable?

Subsumption: structure knowledge, compute taxonomy

- $KB \models C \sqsubseteq D$?

Equivalence: check if two fuzzy concepts are the same

- $KB \models C = D$?

Graded instantiation: Check if individual *a* instance of class *C* to degree at least *r*

- $KB \models \langle a{:}C, r\rangle$?

BTVB: Best Truth Value Bound problem

- $|a{:}C|_{KB} = \sup\{r \mid KB \models \langle a{:}C, r\rangle\}$?

Top-k retrieval: Retrieve the top-k individuals that instantiate *C* w.r.t. best truth value bound

- $ans_{top-k}(KB, C) = Top_k\{\langle a, v\rangle \mid v = |a{:}C)|_{KB}\}$

Some Notes on . . .

- Value restrictions:
  - In classical DLs, $\forall R.C \equiv \neg\exists R.\neg C$
  - The same is not true, in general, in fuzzy DLs (depends on the operators' semantics, true for Łukasiewicz, but not true in Gödel logic)
  - Is it acceptable that $\forall hasParent.Human \not\equiv \neg\exists hasParent.\neg Human$? Recall that in Ł and Zadeh,

    $\forall x.\phi \equiv \neg\exists x\, \neg\phi$
- Models:
  - In classical DLs $\top \sqsubseteq \neg(\forall R.A) \sqcap (\neg\exists R.\neg A)$ has no classical model
  - In Gödel logic it has no finite model, but has an infinite model
- The choice of the appropriate semantics of the logical connectives is important.
  - Should have reasonable logical properties
  - Certainly it must have efficient algorithms solving basic inference problems
- Łukasiewicz Logic seems the best compromise, though Zadeh semantics has been considered historically in DLs (we recall that Zadeh semantics is not considered by fuzzy logicians)
- For disjointness it is better to use $C \sqcap D \sqsubseteq \bot$ rather than $C \sqsubseteq \neg D$
  - they are not the same, e.g. $A \sqsubseteq \neg A$ says that $A^{\mathcal{I}}(x) \leq 0.5$ holds, for all $\mathcal{I}$ and for all $x \in \Delta^{\mathcal{I}}$

Towards fuzzy OWL Lite and OWL DL

- Recall that OWL Lite and OWL DL relate to $\mathcal{SHIF}(D)$ and $\mathcal{SHOIN}(D)$, respectively
- We need to extend the semantics of fuzzy $\mathcal{ALC}$ to fuzzy $\mathcal{SHOIN}(D) = \mathcal{ALCHOINR}_{+}(D)$
- Additionally, we add
  - modifiers (e.g., *very*)
  - concrete fuzzy concepts (e.g., *Young*)
  - both additions have explicit membership functions

Number Restrictions, Inverse and Transitive roles

- The semantics of the concept $(\geq n\ R)$ is:

$$\exists y_1, \ldots, y_n. \bigwedge_{i=1}^{n} R(x, y_i) \wedge \bigwedge_{1 \leq i < j \leq n} y_i \neq y_j \ .$$

- The semantics of the concept $(\leq n\ R)$ is:

$$(\leq n\ R)^{\mathcal{I}}(x) = \forall y_1, \ldots, y_{n+1}. \bigwedge_{i=1}^{n+1} R(x, y_i) \rightarrow \bigvee_{1 \leq i < j \leq n+1} y_i = y_j \ .$$

- Note: $(\geq 1\ R) \equiv \exists R.\top$
- For inverse roles we have for all $x, y \in \Delta^{\mathcal{I}}$

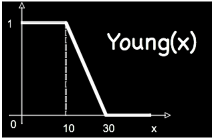
$$R^{\mathcal{I}}(x, y) = R^{\mathcal{I}}(y, x)$$

- For transitive roles R we impose: for all $x, y \in \Delta^{\mathcal{I}}$

$$R^{\mathcal{I}}(x, y) \geq \sup_{z \in \Delta^{\mathcal{I}}} \min(R^{\mathcal{I}}(x, z), R^{\mathcal{I}}(z, y))$$

Concrete fuzzy concepts

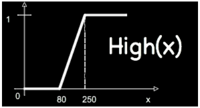
- E.g., *Small*, *Young*, *High*, *etc*. with explicit membership function
- Use the idea of concrete domains:
  - $D = \langle \Delta_D, \Phi_D \rangle$
  - Δ_D is an interpretation domain
  - Φ_D is the set of concrete fuzzy domain predicates d with a predefined arity $n = 1, 2$ and fixed interpretation $d^D : \Delta_D^n \to [0, 1]$
  - For instance,



$$\begin{array}{rcl} Minor & = & Person \sqcap \exists hasAge. \leq_{18} \\ YoungPerson & = & Person \sqcap \exists hasAge. Young \\ & & functional(hasAge) \end{array}$$

Modifiers

- *Very*, *moreOrLess*, *slightly*, etc.
- Apply to fuzzy sets to change their membership function
  - $\mathit{very}(x) = x^2$
  - $\mathit{slightly}(x) = \sqrt{x}$
- For instance,



$$\mathit{SportsCar} = \mathit{Car} \sqcap \exists \mathit{speed}.\mathit{very}(\mathit{High})$$

Fuzzy $\mathcal{SHOIN}(D)$

Concepts:

		Syntax	Semantics
C, D	$\longrightarrow$	$\top$ |	$\top(x)$
		$\bot$ |	$\bot\ (x)$
		A |	$A(x)$
		$(C \sqcap D)$ |	$C_1(x) \wedge C_2(x)$
		$(C \sqcup D)$ |	$C_1(x) \vee C_2(x)$
		$(\neg C)$ |	$\neg C(x)$
		$(\exists R.C)$ |	$\exists x\ R(x,y) \wedge C(y)$
		$(\forall R.C)$ |	$\forall x\ R(x,y) \rightarrow C(y)$
		$\{a\}$ |	$x = a$
		$(\geq n\ R)$ |	$\exists y_1, \ldots, y_n. \bigwedge_{i=1}^{n} R(x, y_i) \wedge \bigwedge_{1 \leq i < j \leq n} y_i \neq y_j$
		$(\leq n\ R)$ |	$\forall y_1, \ldots, y_{n+1}. \bigwedge_{i=1}^{n+1} R(x, y_i) \rightarrow \bigvee_{1 \leq i < j \leq n+1} y_i = y_j$
		FCC |	$\mu_{FCC}(x)$
		$M(C)$ |	$\mu_M(C(x))$
R	$\longrightarrow$	P |	$P(x,y)$
		P^- |	$P(y,x)$

Assertions:

		Syntax	Semantics
α	$\longrightarrow$	$\langle a{:}C, r\rangle$ |	$r \rightarrow C(a)$
		$\langle (a,b){:}R, r\rangle$	$r \rightarrow R(a,b)$

Axioms:

		Syntax	Semantics
τ	$\longrightarrow$	$\langle C \sqsubseteq D, r\rangle$ |	$\forall x\ r \rightarrow (C(x) \rightarrow D(x))$, where $\rightarrow$ is r-implication
		$fun(R)$ |	$\forall x \forall y \forall z\ R(x,y) \wedge R(x,z) \rightarrow y = z$
		$trans(R)$	$(\exists z\ R(x,z) \wedge R(z,y)) \rightarrow R(x,y)$

Example (Graded Entailment)

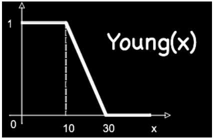
		
audi_tt	*mg*	*ferrari_enzo*

Car	*speed*
audi_tt	243
mg	≤ 170
ferrari_enzo	≥ 350

$$SportsCar = Car \sqcap \exists hasSpeed.very(High)$$

$$\begin{array}{lcl} KB & \models & \langle ferrari\_enzo{:}SportsCar, 1\rangle \\ KB & \models & \langle audi\_tt{:}SportsCar, 0.92\rangle \\ KB & \models & \langle mg{:}\neg SportsCar, 0.72\rangle \end{array}$$

Example (Graded Subsumption)

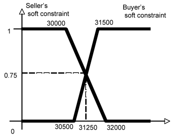


$$\begin{array}{rcl} Minor & = & Person \sqcap \exists hasAge. \leq_{18} \\ YoungPerson & = & Person \sqcap \exists hasAge. Young \end{array}$$

$$KB \models \langle Minor \sqsubseteq YoungPerson, 0.2 \rangle$$

Note: without an explicit membership function of *Young*, this inference cannot be drawn

Example (Simplified Negotiation)



- a car seller sells an Audi TT for 31500 €, as from the catalog price.
- a buyer is looking for a sports-car, but wants to to pay not more than around 30000 €
- classical DLs: the problem relies on the crisp conditions on price
- more fine grained approach: to consider prices as fuzzy sets (as usual in negotiation)
  - seller may consider optimal to sell above 31500 €, but can go down to 30500 €
  - the buyer prefers to spend less than 30000 €, but can go up to 32000 €

$$\begin{array}{lcl} AudiTT & = & SportsCar \sqcap \exists hasPrice.R(x; 30500, 31500) \\ Query & = & SportsCar \sqcap \exists hasPrice.L(x; 30000, 32000) \end{array}$$

  - highest degree to which the concept
    $C = AudiTT \sqcap Query$
    is satisfiable is 0.75 (the possibility that the Audi TT and the query matches is 0.75)
  - the car may be sold at 31250 €

Reasoning [19, 17, 18]

Depends on the semantics and reasoning method (tableau-based or MILP-based)

Tableaux method: under Zadeh semantics

- a tableau exists for fuzzy $\mathcal{SHIN}$, solving the satisfiability problem
- classical blocking methods apply similarly in the fuzzy variant
- the management of General concept inclusions (GCI's) is more complicated compared to the crisp case
- a translation of fuzzy $\mathcal{SHOIN}$ to crisp $\mathcal{SHOIN}$ also exists (not addressed here)
- the tableaux method is not suitable to deal with fuzzy concrete concepts and modifiers
- the BTVB can be solved, but not efficiently

MILP based method: under Zadeh semantics, Łukasiewicz semantics, and classical semantics

- exists for fuzzy $\mathcal{ALC}$ + linear modifiers + fuzzy concrete concepts [20, 21, 2]
- exists for fuzzy $\mathcal{SHIF}$ + linear modifiers + fuzzy concrete concepts (implemented in fuzzyDL reasoner, but not published yet [1, 2])
- solves the BTVB as primary problem

MIQP based method: using Mixed Integer Quadratically Constrained Programming optimization problem (MICQP) for product T-norm

- exists for fuzzy $\mathcal{SHIF}$ + linear modifiers + fuzzy concrete concepts (implemented in fuzzyDL reasoner, but not published yet [1]). Important as it simulates probabilistic reasoning under independent event assumption.
- solves the BTVB as primary problem
- the fuzzyDL solver also allows to mix all three semantics

Problem with fuzzy tableau

- Usual fuzzy tableaux calculus does not work anymore with
  - modifiers and concrete fuzzy concepts
  - Łukasiewicz Logic
  - Product T-norm
- Usual fuzzy tableaux calculus does not solve the BTVB problem
- New algorithm uses bounded Mixed Integer Programming oracle, as for Many Valued Logics
  - Recall: the *general MILP problem* is to find

$$\begin{array}{l} \bar{\mathbf{x}} \in \mathbb{Q}^k, \bar{\mathbf{y}} \in \mathbb{Z}^m \\ f(\bar{\mathbf{x}}, \bar{\mathbf{y}}) = \min\{f(\mathbf{x}, \mathbf{y}) : A\mathbf{x} + B\mathbf{y} \geq \mathbf{h}\} \\ A, B \text{ integer matrixes} \end{array}$$

  - $|\varphi|_{KB} = \min\{x \mid KB \cup \{\langle \varphi \leq x \rangle \text{ satisfiable}\}$

$\mathcal{ALC}$ MILP Tableau rules under Zadeh semantics (excerpt)

$x \bullet \{\langle C_1 \sqcap C_2, \geq, l\rangle, \ldots\}$	$\longrightarrow_{\sqcap}$	$x \bullet \{\langle C_1 \sqcap C_2, \geq, l\rangle, \langle C_1, \geq, l\rangle, \langle C_2, \geq, l\rangle, \ldots\}$
$x \bullet \{\langle C_1 \sqcup C_2, \geq, l\rangle, \ldots\}$	$\longrightarrow_{\sqcup}$	$x \bullet \{\langle C_1 \sqcup C_2, \geq, l\rangle, \langle C_1, \geq, x_1\rangle, \langle C_2, \geq, x_2\rangle,$ $x_1 + x_2 = l, x_1 \leq y, x_2 \leq 1 - y,$ $x_i \in [0, 1], y \in \{0, 1\}, \ldots\}$
$x \bullet \{\langle \exists R.C, \geq, l\rangle, \ldots\}$	$\longrightarrow_{\exists}$	$x \bullet \{\langle \exists R.C, \geq, l\rangle, \ldots\}$ $\langle R, \geq, l\rangle \downarrow$ $y \bullet \{\langle C, \geq, l\rangle\}$
$x \bullet \{\langle \forall R.C, \geq, l_1\rangle, \ldots\}$ $\langle R, \geq, l_2\rangle \downarrow$ $y \bullet \{\ldots\}$	$\longrightarrow_{\forall}$	$x \bullet \{\langle \forall R.C, \geq, l_1\rangle, \ldots\}$ $\langle R, \geq, l_2\rangle \downarrow$ $y \bullet \{\ldots, \langle C, \geq, x\rangle$ $x + y \geq l_1, x \leq y, l_1 + l_2 \leq 2 - y,$ $x \in [0, 1], y \in \{0, 1\}\}$
$x \bullet \{A \sqsubseteq C, \langle A, \geq, l\rangle, \ldots\}$	$\longrightarrow_{\sqsubseteq_1}$	$x \bullet \{A \sqsubseteq C, \langle C, \geq, l\rangle, \ldots\}$
$x \bullet \{C \sqsubseteq A, \langle A, \leq, l\rangle, \ldots\}$	$\longrightarrow_{\sqsubseteq_2}$	$x \bullet \{C \sqsubseteq A, \langle C, \leq, l\rangle, \ldots\}$
$x \bullet \{C \sqsubseteq D, \ldots\}$	$\longrightarrow_{\sqsubseteq}$	$x \bullet \{C \sqsubseteq D, \langle C, \leq, x\rangle, \langle D, \geq, x\rangle, x \in [0, 1], \ldots\}$
$x \bullet \{\langle ls(k_1, k_2, a, b), \geq, l\rangle, \ldots\}$	$\longrightarrow_{\sqsubseteq}$	$x \bullet \{ls(k_1, k_2, a, b), y_1 + y_2 + y_3 = 1, y_i \in \{0, 1\},$ $x + (k_2 - a) \cdot y_1 \leq k_2, x + (k_1 - a) \cdot y_2 \geq k_1,$ $x + (k_2 - b) \cdot y_2 \geq k_2,$ $x + (b - a) \cdot l + (k_2 - a) \cdot y_2 \leq k_2 - a + b,$ $x + (k_1 - b) \cdot y_3 \leq k_1, l + y_3 \leq 1, \ldots\}$

Example

- Suppose
$$KB = \begin{cases} A \sqcap B \sqsubseteq C \\ \langle a{:}A \geq 0.3 \rangle \\ \langle a{:}B \geq 0.4 \rangle \end{cases}$$
$$Query \quad := \quad |a{:}C|_{KB} = \min\{x \mid KB \cup \{\langle a{:}C \leq x \rangle \text{ satisfiable}\}$$

Step	Tree	
1.	$a \bullet \{\langle A, \geq, 0.3 \rangle, \langle B, \geq, 0.4 \rangle, \langle C, \leq, x \rangle\}$	(Hypothesis)
2.	$\cup\{\langle A \sqcap B, \leq, x \rangle\}$	$(\rightarrow_{\sqsubseteq_2})$
3.	$\cup\{\langle A, \leq, x_1 \rangle, \langle B, \leq, x_2 \rangle\}$ $\cup\{x = x_1 + x_2 - 1, 1 - y \leq x_1, y \leq x_2\}$ $\cup\{x_i \in [0, 1], y \in \{0, 1\}\}$	$(\rightarrow_{\sqcap_{\leq}})$
4.	find $\min\{x \mid \langle a{:}A \geq 0.3 \rangle, \langle a{:}B \geq 0.4 \rangle,$ $\langle a{:}C \leq x \rangle, \langle a{:}A \leq x_1 \rangle, \langle a{:}B \leq x_2 \rangle,$ $x = x_1 + x_2 - 1, 1 - y \leq x_1, y \leq x_2,$ $x_i \in [0, 1], y \in \{0, 1\}\}$	(MILP Oracle)
5.	MILP oracle: $\mathbf{x} = \mathbf{0.3}$	

Implementation issues

- Several options exists:
  - Try to map fuzzy DLs to classical DLs
    - difficult to work with modifiers and concrete fuzzy concepts
  - Try to map fuzzy DLs to some fuzzy logic programming framework
    - A lot of work exists about mappings among classical DLs and LPs
    - But, needs a theorem prover for fuzzy LPs
  - Build an ad-hoc theorem prover for fuzzy DLs, using e.g., MILP
- A theorem prover for fuzzy $\mathcal{SHIF}$ + linear hedges + concrete fuzzy concepts + linear equational constraints + datatypes, under classical, Zadeh, Lukasiewicz and Product t-norm semantics has been implemented (http://gaia.isti.cnr.it/~straccia)
- FIRE: a fuzzy DL theorem prover for fuzzy $\mathcal{SHIN}$ under Zadeh semantics (http://www.image.ece.ntua.gr/~nsimou/)

Top-k retrieval in tractable DLs: the case of DL-Lite/DLR-Lite [25, 30]

- DL-Lite/DLR-Lite [3]: a simple, but interesting DLs
- Captures important subset of UML/ER diagrams
- Computationally tractable DL to query large databases
- Sub-linear, i.e. LOGSpace in data complexity
  - (same cost as for SQL)
- Good for very large database tables, with limited declarative schema design w.r.t. OWL-DL

- Knowledge base: $KB = \langle \mathcal{T}, \mathcal{A} \rangle$, where $\mathcal{T}$ and $\mathcal{A}$ are finite sets of axioms and assertions
- Axiom: $Cl \sqsubseteq Cr$ (inclusion axiom)
- Note for inclusion axioms: the language for left hand side is different from the one for right hand side
- DL-Lite$_{core}$:
  - Concepts: $$\begin{array}{lcl} Cl & \to & A \mid \exists R \\ Cr & \to & A \mid \exists R \mid \neg A \mid \neg \exists R \\ R & \to & P \mid P^{-} \end{array}$$
  - Assertion: $a{:}A$, $(a, b){:}P$
- DLR-Lite$_{core}$: (n-ary roles)
  - Concepts: $$\begin{array}{lcl} Cl & \to & A \mid \exists P[i] \\ Cr & \to & A \mid \exists P[i] \mid \neg A \mid \neg \exists P[i] \end{array}$$
  - $\exists P[i]$ is the projection on i-th column
  - Assertion: $a{:}A$, $\langle a_1, \ldots, a_n \rangle{:}P$
- Assertions are stored in relational tables
- Conjunctive query: $q(\mathbf{x}) \leftarrow \exists \mathbf{y}.conj(\mathbf{x}, \mathbf{y})$
  $conj$ is an aggregation of expressions of the form $B(z)$ or $P(z_1, z_2)$,

Top-k retrieval in DL-Lite/DLR-Lite

- We extend the query formalism: conjunctive queries, where fuzzy predicates may appear
- conjunctive query

$$q(\mathbf{x}, s) \leftarrow \exists \mathbf{y}.conj(\mathbf{x}, \mathbf{y}), s = f(p_1(\mathbf{z}_1), \ldots, p_n(\mathbf{z}_n))$$

1. $\mathbf{x}$ are the *distinguished variables*;
2. s is the *score variable*, taking values in $[0, 1]$;
3. $\mathbf{y}$ are existentially quantified variables, called *non-distinguished variables*;
4. $conj(\mathbf{x}, \mathbf{y})$ is a conjunction of DL-Lite/DLR-Lite atoms $R(\mathbf{z})$ in KB;
5. $\mathbf{z}$ are tuples of constants in KB or variables in $\mathbf{x}$ or $\mathbf{y}$;
6. $\mathbf{z_i}$ are tuples of constants in KB or variables in $\mathbf{x}$ or $\mathbf{y}$;
7. p_i is an n_i-ary *fuzzy predicate* assigning to each n_i-ary tuple $\mathbf{c}_i$ the *score* $p_i(\mathbf{c}_i) \in [0, 1]$;
8. f is a monotone *scoring* function $f\colon [0, 1]^n \rightarrow [0, 1]$, which combines the scores of the n fuzzy predicates $p_i(\mathbf{c}_i)$

Example:

$$
\begin{array}{rcl}
\mathit{Hotel} & \sqsubseteq & \exists \mathit{HasHLoc} \\
\mathit{Hotel} & \sqsubseteq & \exists \mathit{HasHPrice} \\
\mathit{Conference} & \sqsubseteq & \exists \mathit{HasCLoc} \\
\mathit{Hotel} & \sqsubseteq & \neg \mathit{Conference}
\end{array}
$$

HasHLoc		*HasCLoc*		*HasHPrice*	
HotelID	*HasLoc*	*ConfID*	*HasLoc*	*HotelID*	*Price*
$h1$	$hl1$	$c1$	$cl1$	$h1$	150
$h2$	$hl2$	$c2$	$cl2$	$h2$	200
⋮	⋮	⋮	⋮	⋮	⋮

$$
\begin{array}{c}
q(h, s) \leftarrow \mathit{HasHLoc}(h, hl), \mathit{HasHPrice}(h, p), \mathit{Distance}(hl, cl, d) \\
\mathit{HasCLoc}(c1, cl), s = \mathit{cheap}(p) \cdot \mathit{close}(d) \;.
\end{array}
$$

where the fuzzy predicates *cheap* and *close* are defined as

$$
\begin{array}{rcl}
\mathit{close}(d) & = & \mathit{ls}(0, 2km, d) \\
\mathit{cheap}(p) & = & \mathit{ls}(0, 300, p)
\end{array}
$$

Semantics informally:

- a conjunctive query
  $$q(\mathbf{x}, s) \leftarrow \exists \mathbf{y}.conj(\mathbf{x}, \mathbf{y}), s = f(p_1(\mathbf{z}_1), \ldots, p_n(\mathbf{z}_n))$$
  is interpreted in an interpretation $\mathcal{I}$ as the set
  $$q^{\mathcal{I}} = \{\langle \mathbf{c}, v\rangle \in \Delta \times \ldots \times \Delta \times [0,1] \mid \ldots$$
  such that when we consider the substitution
  $$\theta = \{\mathbf{x}/\mathbf{c}, s/v\}$$
  the formula
  $$\exists \mathbf{y}.conj(\mathbf{x}, \mathbf{y}) \wedge s = f(p_1(\mathbf{z}_1), \ldots, p_n(\mathbf{z}_n))$$
  evaluates to true in $\mathcal{I}$.
- Model of a query: $\mathcal{I} \models q(\mathbf{c}, v)$ iff $\langle \mathbf{c}, v\rangle \in q^{\mathcal{I}}$
- Entailment: $KB \models q(\mathbf{c}, v)$ iff $\mathcal{I} \models KB$ implies $\mathcal{I} \models q(\mathbf{c}, v)$
- Top-k retrieval: $ans_{top-k}(KB, q) = Top_k\{\langle \mathbf{c}, v\rangle \mid KB \models q(\mathbf{c}, v)\}$

How to determine the top-k answers of a query?

- Overall strategy: three steps
  1. Check if KB is satisfiable, as querying a non-satisfiable KB is meaningless (checkable in linear time)
  2. Query q is *reformulated* into a set of conjunctive queries $r(q, \mathcal{T})$
     - Basic idea: reformulation procedure closely resembles a top-down resolution procedure for logic programming

$$
\begin{array}{rcl}
q(x, s) & \leftarrow & B(x), A(x), s = f(x) \\
B_1 & \sqsubseteq & A \\
B_2 & \sqsubseteq & A \\
\hline\hline
q(x, s) & \leftarrow & B(x), B_1(x), s = f(x) \\
q(x, s) & \leftarrow & B(x), B_2(x), s = f(x)
\end{array}
$$

  3. The reformulated queries in $r(q, \mathcal{T})$ are evaluated over $\mathcal{A}$ (seen as a database) using standard top-k techniques for DBs
     - for all $q_i \in r(q, \mathcal{T})$, $ans_{top-k}(q_i, \mathcal{A})$ = top-k SQL query over $\mathcal{A}$ database
     - $ans_{top-k}(KB, q) = Top_k(\bigcup_{q_i \in r(q, \mathcal{T})} ans_k(q_i, \mathcal{A}))$

Small Example:

P_2	
0	s
3	t
4	q
6	q

B
1
2
5
7

$$\mathcal{T} = \{\exists P_2^- \sqsubseteq A, A \sqsubseteq \exists P_1, B \sqsubseteq \exists P_2\}$$

$$q(x,s) \leftarrow P_2(x,y), P_1(y,z), s = \max(0, 1 - x/10)$$

$$q(x,s) \leftarrow P_2(x,y), A(y), s = \max(0, 1 - x/10)$$
$$q(x,s) \leftarrow P_2(x,y), P_2(z,y), s = \max(0, 1 - x/10)$$
$$q(x,s) \leftarrow P_2(x,y), s = \max(0, 1 - x/10)$$
$$q(x,s) \leftarrow B(x), s = \max(0, 1 - x/10)$$

$$q_1(x,s) \leftarrow P_2(x,y), s = \max(0, 1 - x/10)$$
$$q_2(x,s) \leftarrow B(x), s = \max(0, 1 - x/10)$$

$$ans_{top-3}(\mathcal{A}, q_1) = [\langle 0, 1.0\rangle, \langle 3, 0.7\rangle, \langle 4, 0.6\rangle]$$
$$ans_{top-3}(\mathcal{A}, q_2) = [\langle 1, 0.9\rangle, \langle 2, 0.8\rangle, \langle 5, 0.5\rangle]$$

$$ans_{top-k}(KB, q) = [\langle 0, 1.0\rangle, \langle 1, 0.9\rangle, \langle 2, 0.8\rangle]$$

Proposition

Given a DL-Lite KB $KB = \langle \mathcal{T}, \mathcal{A}\rangle$ and a query q then we can compute $ans_{top-k}(KB, q)$ in (sub) linear time w.r.t. the size of $\mathcal{A}$. The same holds for the description logic DLR-Lite.

Tool exists and implemented in the DLMedia system

`http://gaia.isti.cnr.it/~straccia`

DLMedia: a Multimedia Information Retrieval System [33]

- Based on fuzzy DLR-Lite with similarity predicates
  - Axioms: $Rl_1 \sqcap \ldots \sqcap Rl_m \sqsubseteq Rr$

$$
\begin{array}{lcl}
Rr & \longrightarrow & A \mid \exists[i_1, \ldots, i_k]R \\
Rl & \longrightarrow & A \mid \exists[i_1, \ldots, i_k]R \mid \exists[i_1, \ldots, i_k]R.(Cond_1 \sqcap \ldots \sqcap Cond_l) \\
Cond & \longrightarrow & ([i] \leq v) \mid ([i] < v) \mid ([i] \geq v) \mid ([i] > v) \mid ([i] = v) \mid ([i] \neq v) \mid \\
 & & ([i]\ simTxt\ 'k_1, \ldots, k_n') \mid ([i]\ simImg\ URN)
\end{array}
$$

  - $\exists[i_1, \ldots, i_k]R$ is the projection of the relation R on the columns $i_1, \ldots, i_k$
  - $\exists[i_1, \ldots, i_k]R.(Cond_1 \sqcap \ldots \sqcap Cond_l)$ further restricts the projection $\exists[i_1, \ldots, i_k]R$ according to the conditions specified in $Cond_i$
  - $([i]\ simTxt\ 'k_1 \ldots k_n')$ evaluates the degree of being the text of the i-th column similar to the list of keywords $k_1 \ldots k_n$
  - $([i]\ simImg\ URN)$ returns the system's degree of being the image identified by the i-th column similar to the image identified by the URN
  - Facts: $\langle R(c_1, \ldots, c_n), s\rangle$

- Example axioms

$$
\begin{array}{l}
\exists[1,2]Person \sqsubseteq \exists[1,2]hasAge \\
\qquad\qquad\text{// constrains relation } hasAge(name, age) \\
\exists[3,1]Person \sqsubseteq \exists[1,2]hasChild \\
\qquad\qquad\text{// constrains relation } hasChild(father\_name, name) \\
\exists[4,1]Person \sqsubseteq \exists[1,2]hasChild \\
\qquad\qquad\text{// constrains relation } hasChild(mother\_name, name) \\
\exists[3,1]Person.(([2] \geq 18) \sqcap ([5] =' female')) \sqsubseteq \exists[1,2]hasAdultDaughter \\
\qquad\qquad\text{// constrains relation } hasAdultDaughter(father\_name, name)
\end{array}
$$

- On the other hand examples axioms involving similarity predicates are,

$$\exists[1]ImageDescr.([2]\ simImg\ urn1) \sqsubseteq Child \quad (1)$$

$$\exists[1]Title.([2]\ simTxt\ 'lion') \sqsubseteq Lion \quad (2)$$

where *urn*1 identifies the image

- Example queries

$q(x) \leftarrow Child(x)$
// find objects about a child (strictly speaking, find instances of *Child*)

$q(x) \leftarrow CreatorName(x, y) \wedge (y =' paolo'), Title(x, z), (z\ simTxt\ ' tour')$
// find images made by Paolo whose title is about 'tour'

$q(x) \leftarrow ImageDescr(x, y) \wedge (y\ simImg\ urn2)$
// find images similar to a given image identified by *urn*2

$q(x) \leftarrow ImageObject(x) \wedge isAbout(x, y_1) \wedge Car(y_1) \wedge isAbout(x, y_2) \wedge Racing(y_2)$
// find image objects about cars racing

Fuzzy LPs Basics [4, 6, 7, 22, 23, 29, 35]

- Many Logic Programming (LP) frameworks have been proposed to manage uncertain and imprecise information. They differ in:
  - The underlying notion of uncertainty and vagueness: probability, possibility, many-valued, fuzzy logics
  - How values, associated to rules and facts, are managed
- We consider fuzzy LPs, where
  - Truth space is $[0, 1]$
  - Interpretation is a mapping $I : B_{\mathcal{P}} \to [0, 1]$
  - Generalized LP rules are of the form

$$R(\mathbf{x}) \leftarrow \exists \mathbf{y}. f(R_1(\mathbf{z}_1), \ldots, R_l(\mathbf{z}_l), p_1(\mathbf{z}'_1), \ldots, p_h(\mathbf{z}'_h)) \ ,$$

    - Meaning of rules: "take the truth-values of all $R_i(\mathbf{z}_i), p_j(\mathbf{z}'_j)$, combine them using the truth combination function f, and assign the result to $R(\mathbf{x})$"

- Same meaning as for fuzzy DLR-Lite queries

$$R(\mathbf{x}, s) \leftarrow \exists \mathbf{y}.conj(\mathbf{x}, \mathbf{y}), s = f(p_1(\mathbf{z}_1), \ldots, p_{l+h}(\mathbf{z}_{l+h}))$$

1. $\mathbf{x}$ are the *distinguished variables*;
2. s is the *score variable*, taking values in $[0, 1]$;
3. $\mathbf{y}$ are existentially quantified variables, called *non-distinguished variables*;
4. $conj(\mathbf{x}, \mathbf{y})$ is a list of atoms $R_i(\mathbf{z})$ in KB;
5. $\mathbf{z}$ are tuples of constants in KB or variables in $\mathbf{x}$ or $\mathbf{y}$;
6. $\mathbf{z_i}$ are tuples of constants in KB or variables in $\mathbf{x}$ or $\mathbf{y}$;
7. p_i is an n_i-ary *fuzzy predicate* assigning to each n_i-ary tuple $\mathbf{c}_i$ the *score* $p_i(\mathbf{c}_i) \in [0, 1]$;
8. f is a monotone *scoring* function $f\colon [0, 1]^{l+h} \to [0, 1]$, which combines the scores of the n fuzzy predicates $p_i(\mathbf{c}_i)$

Semantics of fuzzy LPs

- Model of a LP:

$$\begin{array}{lll} I \models \mathcal{P} & \text{iff} & I \models r, \text{ for all } r \in \mathcal{P}^* \\ I \models A \leftarrow \varphi & \text{iff} & I(\varphi) \leq I(A) \end{array}$$

- Least model exists and is least fixed-point of

$$T_{\mathcal{P}}(I)(A) = I(\varphi)$$

  for all $A \leftarrow \varphi \in \mathcal{P}^*$

- Fuzzy LPs may be tricky:

$$\begin{array}{lll} \langle A, 0 \rangle & & \\ A & \leftarrow & (A+1)/2 \end{array}$$

  In the minimal model the truth of A is 1 (requires ω $T_{\mathcal{P}}$ iterations)!

- There are several ways to avoid this pathological behavior:
  - We consider $\mathcal{T} = \{0, \frac{1}{n}, \frac{2}{n} \ldots, \frac{n-1}{n}, 1\}$, n natural number, e.g. $n = 100$

Example: Soft shopping agent

- I may represent my preferences in Logic Programming with the rules

$$
\begin{array}{rcl}
Pref_1(x, p, s) & \leftarrow & HasPrice(x, p), LS(10000, 14000, p, s) \\
Pref_2(x, s) & \leftarrow & HasKM(x, k), LS(13000, 17000, k, s) \\
Buy(x, p, s) & \leftarrow & Pref_1(x, p, s_1), Pref_2(x, s_2), s = 0.7 \cdot s_1 + 0.3 \cdot s_2
\end{array}
$$

ID	MODEL	PRICE	KM
455	MAZDA 3	12500	10000
34	ALFA 156	12000	15000
1812	FORD FOCUS	11000	16000
⋮	⋮	⋮	⋮

- Problem: All tuples of the database have a score:
  - We cannot compute the score of all tuples, then rank them. Brute force approach not feasible.
- Top-k problem: Determine efficiently just the top-k ranked tuples, without evaluating the score of all tuples. E.g. top-3 tuples

ID	PRICE	SCORE
1812	11000	0.6
455	12500	0.56
34	12000	0.50

Top-k retrieval in LPs

- If the database contains a huge amount of facts, a brute force approach fails:
  - one cannot anymore compute the score of all tuples, rank all of them and only then return the top-k
- Better solutions exists for restricted fuzzy LP languages: Datalog + restriction on the score combination functions appearing in the body [29, 32]

Basic Idea

- We do not compute all answers, but determine answers incrementally
- At each step i, from the tuples seen so far in the database, we compute a threshold δ
- The threshold δ has the property that any successively retrieved answer will have a score $s \leq \delta$
- Therefore, we can stop as soon as we have gathered k answers above δ, because any successively computed answer will have a score below δ

Procedure *TopAnswers*($\mathcal{K}$, Q, k)
Input: KB $\mathcal{K}$, intensional query relation symbol Q, $k \geq 1$;
Output: Mapping $\texttt{rankedList}$ such that $\texttt{rankedList}(Q)$ contains top-k answers of Q
Init: $\delta = 1$, **for all** rules $r : P(\mathbf{x}) \leftarrow \phi$ in P **do**
  if P intensional **then** $\texttt{rankedList}(P) = \emptyset$;
  if P extensional **then** $\texttt{rankedList}(P) = T_P$ **endfor**
1. **loop**
2. $\texttt{Active} := \{Q\}$, $\texttt{dg} := \{Q\}$, $\texttt{in} := \emptyset$,
  for all rules $r : P(\mathbf{x}) \leftarrow \phi$ **do** $\texttt{exp}(P, r) = \texttt{false}$;
3. **while** ($\texttt{Active} \neq \emptyset$) **do**
4. **select** $P \in \texttt{A}$ where $r : P(\mathbf{x}) \leftarrow \phi$, $\texttt{Active} := \texttt{Active} \setminus \{P\}$, $\texttt{dg} := \texttt{dg} \cup \texttt{s}(P, r)$;
5. $\langle \mathbf{t}, s \rangle :=$ *getNextTuple*(P, r)
6. **if** $\langle \mathbf{t}, s \rangle \neq \texttt{NULL}$ **then** insert $\langle \mathbf{t}, s \rangle$ into $\texttt{rankedList}(P)$,
  $\texttt{Active} := \texttt{Active} \cup (\texttt{p}(P) \cap \texttt{dg})$;
7. **if not** $\texttt{exp}(P, r)$ **then** $\texttt{exp}(P, r) = \texttt{true}$,
  $\texttt{Active} := \texttt{Active} \cup (\texttt{s}(P, r) \setminus \texttt{in})$, $\texttt{in} := \texttt{in} \cup \texttt{s}(p, r)$;
  endwhile
8. Update threshold δ;
9. **until** ($\texttt{rankedList}(Q)$ does contain k top-ranked tuples with score above δ)
  or ($\texttt{rL}' = \texttt{rankedList}$);
10. **return** top-k ranked tuples in $\texttt{rankedList}(Q)$;

Procedure *getNextTuple*(P, r)
Input: intensional relation symbol P and rule $r : P(\mathbf{x}) \leftarrow \exists \mathbf{y}.f(R_1(\mathbf{z}_1), \ldots, R_n(\mathbf{z}_l)) \in P$;
Output: Next tuple satisfying the body of the r together with the score
Init:

```
        loop
1.          Generate next new instance tuple ⟨t, s⟩ of P, using tuples in rankedList(R_i) and RankSQL
2.          if there is no ⟨t, s'⟩ ∈ rankedList(P, r) with s ≤ s' then exit loop
        until no new valid join tuple can be generated
3.      return ⟨t, s⟩ if it exists else return NULL
```

Example

Logic Program $\mathcal{P}$ is

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

RecordID	r_1			r_2		
1	a	b	1.0	m	h	0.95
2	c	d	0.9	m	j	0.85
3	e	f	0.8	f	k	0.75
4	l	m	0.7	m	n	0.65
5	o	p	0.6	p	q	0.55
⋮	⋮	⋮	⋮	⋮	⋮	⋮

What is

$$Top_1(\mathcal{P}, q) = Top_1\{\langle c, s\rangle \mid \mathcal{P} \models q(c, s)\} \ ?$$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

RecordID	r_1			r_2		
1	*a*	*b*	1.0	*m*	*h*	0.95
2	*c*	*d*	0.9	*m*	*j*	0.85
3	*e*	*f*	0.8	*f*	*k*	0.75
4	*l*	*m*	0.7	*m*	*n*	0.65
5	*o*	*p*	0.6	*p*	*q*	0.55
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Queue	δ
q	1

Predicate	*Answers*
q	$\emptyset$
p	$\emptyset$

$$q(x,s) \leftarrow p(x,s_1), s = s_1$$
$$p(x,s) \leftarrow r_1(x,y,s_1), r_2(y,z,s_2), s = \min(s_1,s_2)$$

RecordID	r_1			r_2		
1	a	b	1.0	m	h	0.95
2	c	d	0.9	m	j	0.85
3	e	f	0.8	f	k	0.75
4	l	m	0.7	m	n	0.65
5	o	p	0.6	p	q	0.55
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Action: select next predicate in queue

Queue	δ
q	1

Predicate	*Answers*
q	$\emptyset$
p	$\emptyset$

$q(x, s) \leftarrow p(x, s_1), s = s_1$
$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$

RecordID	r_1			r_2		
1	a	b	1.0	m	h	0.95
2	c	d	0.9	m	j	0.85
3	e	f	0.8	f	k	0.75
4	l	m	0.7	m	n	0.65
5	o	p	0.6	p	q	0.55
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Action: get next tuple for q

Queue	δ
—	1

Predicate	Answers
q	$\emptyset$
p	$\emptyset$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

RecordID	r_1			r_2		
1	a	b	1.0	m	h	0.95
2	c	d	0.9	m	j	0.85
3	e	f	0.8	f	k	0.75
4	l	m	0.7	m	n	0.65
5	o	p	0.6	p	q	0.55
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Action: no answer yet for q, put all predicates in rule body of q in queue

Queue	δ
p	1

Predicate	*Answers*
q	$\emptyset$
p	$\emptyset$

$$q(x,s) \leftarrow p(x,s_1), s = s_1$$
$$p(x,s) \leftarrow r_1(x,y,s_1), r_2(y,z,s_2), s = \min(s_1,s_2)$$

RecordID	r_1			r_2		
1	a	b	1.0	m	h	0.95
2	c	d	0.9	m	j	0.85
3	e	f	0.8	f	k	0.75
4	l	m	0.7	m	n	0.65
5	o	p	0.6	p	q	0.55
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Action: select next predicate in queue

Queue	δ
p	1

Predicate	*Answers*
q	$\emptyset$
p	$\emptyset$

$q(x, s) \leftarrow p(x, s_1), s = s_1$
$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$

RecordID	r_1			r_2		
1	a	b	1.0	m	h	0.95
2	c	d	0.9	m	j	0.85
3	e	f	0.8	f	k	0.75
4	l	m	0.7	m	n	0.65
5	o	p	0.6	p	q	0.55
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Action: get next tuple for p

Queue	δ
—	1

Predicate	Answers
q	$\emptyset$
p	$\emptyset$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
→	1	*a*	*b*	1.0	*m*	*h*	0.95	←
	2	*c*	*d*	0.9	*m*	*j*	0.85	
	3	*e*	*f*	0.8	*f*	*k*	0.75	
	4	*l*	*m*	0.7	*m*	*n*	0.65	
	5	*o*	*p*	0.6	*p*	*q*	0.55	
	⋮	⋮	⋮	⋮	⋮	⋮	⋮	

Action: get next tuple for p

Queue	δ
–	1

Predicate	*Answers*
q	$\emptyset$
p	$\emptyset$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
	1	*a*	*b*	1.0	*m*	*h*	0.95	$\leftarrow$
$\rightarrow$	2	*c*	*d*	0.9	*m*	*j*	0.85	
	3	*e*	*f*	0.8	*f*	*k*	0.75	
	4	*l*	*m*	0.7	*m*	*n*	0.65	
	5	*o*	*p*	0.6	*p*	*q*	0.55	
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Action: get next tuple for *p*

Queue	δ
–	1

Predicate	*Answers*
q	$\emptyset$
p	$\emptyset$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
	1	*a*	*b*	1.0	*m*	*h*	0.95	
$\rightarrow$	2	*c*	*d*	0.9	*m*	*j*	0.85	$\leftarrow$
	3	*e*	*f*	0.8	*f*	*k*	0.75	
	4	*l*	*m*	0.7	*m*	*n*	0.65	
	5	*o*	*p*	0.6	*p*	*q*	0.55	
	⋮	⋮	⋮	⋮	⋮	⋮	⋮	

Action: get next tuple for *p*

Queue	δ
—	1

Predicate	*Answers*
q	$\emptyset$
p	$\emptyset$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
	1	a	b	1.0	m	h	0.95	
	2	c	d	0.9	m	j	0.85	$\leftarrow$
$\rightarrow$	3	e	f	0.8	f	k	0.75	
	4	l	m	0.7	m	n	0.65	
	5	o	p	0.6	p	q	0.55	
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Action: get next tuple for p

Queue	δ
–	1

Predicate	*Answers*
q	$\emptyset$
p	$\emptyset$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
	1	a	b	1.0	m	h	0.95	
	2	c	d	0.9	m	j	0.85	
$\longrightarrow$	3	e	f	0.8	f	k	0.75	$\longleftarrow$
	4	l	m	0.7	m	n	0.65	
	5	o	p	0.6	p	q	0.55	
	⋮	⋮	⋮	⋮	⋮	⋮	⋮	

Action: get next tuple for p

Queue	δ
—	1

Predicate	*Answers*
q	$\emptyset$
p	$\emptyset$

$$q(x,s) \leftarrow p(x,s_1), s = s_1$$
$$p(x,s) \leftarrow r_1(x,y,s_1), r_2(y,z,s_2), s = \min(s_1,s_2)$$

	RecordID	r_1			r_2			
	1	*a*	*b*	1.0	*m*	*h*	0.95	
	2	*c*	*d*	0.9	*m*	*j*	0.85	
$\longrightarrow$	3	*e*	*f*	0.8	*f*	*k*	0.75	$\longleftarrow$
	4	*l*	*m*	0.7	*m*	*n*	0.65	
	5	*o*	*p*	0.6	*p*	*q*	0.55	
	⋮	⋮	⋮	⋮	⋮	⋮	⋮	

Action: get next tuple for *p*

Queue	δ
—	1

Predicate	*Answers*
q	$\emptyset$
p	$\langle e, 0.75\rangle$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
	1	a	b	1.0	m	h	0.95	
	2	c	d	0.9	m	j	0.85	
$\rightarrow$	3	e	f	0.8	f	k	0.75	$\leftarrow$
	4	l	m	0.7	m	n	0.65	
	5	o	p	0.6	p	q	0.55	
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Action: as p changed, update queue with predicates depending on p

Queue	δ
q	1

Predicate	*Answers*
q	$\emptyset$
p	$\langle e, 0.75 \rangle$

$$q(x,s) \leftarrow p(x,s_1), s = s_1$$
$$p(x,s) \leftarrow r_1(x,y,s_1), r_2(y,z,s_2), s = \min(s_1,s_2)$$

	RecordID	r_1			r_2			
	1	a	b	1.0	m	h	0.95	
	2	c	d	0.9	m	j	0.85	
$\rightarrow$	3	e	f	0.8	f	k	0.75	$\leftarrow$
	4	l	m	0.7	m	n	0.65	
	5	o	p	0.6	p	q	0.55	
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Action: select next predicate from queue, and get next tuple for it

Queue	δ
q	1

Predicate	Answers
q	$\langle e, 0.75\rangle$
p	$\langle e, 0.75\rangle$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
	1	a	b	1.0	m	h	0.95	
	2	c	d	0.9	m	j	0.85	
$\rightarrow$	3	e	f	0.8	f	k	0.75	$\leftarrow$
	4	l	m	0.7	m	n	0.65	
	5	o	p	0.6	p	q	0.55	
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Action: as q changed, update queue with predicates depending on q

Queue	δ
–	1

Predicate	*Answers*
q	$\langle e, 0.75\rangle$
p	$\langle e, 0.75\rangle$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
	1	a	b	1.0	m	h	0.95	
	2	c	d	0.9	m	j	0.85	
$\rightarrow$	3	e	f	0.8	f	k	0.75	$\leftarrow$
	4	l	m	0.7	m	n	0.65	
	5	o	p	0.6	p	q	0.55	
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Action: queue empty, so update threshold and re-start ($0.8 = \max(\min(1.0, 0.75), \min(0.8, 0.95))$)

Queue	δ
–	0.8

Predicate	*Answers*
q	$\langle e, 0.75\rangle$
p	$\langle e, 0.75\rangle$

$$q(x,s) \leftarrow p(x,s_1), s = s_1$$
$$p(x,s) \leftarrow r_1(x,y,s_1), r_2(y,z,s_2), s = \min(s_1,s_2)$$

	RecordID	r_1			r_2			
	1	*a*	*b*	1.0	*m*	*h*	0.95	
	2	*c*	*d*	0.9	*m*	*j*	0.85	
$\rightarrow$	3	*e*	*f*	0.8	*f*	*k*	0.75	$\leftarrow$
	4	*l*	*m*	0.7	*m*	*n*	0.65	
	5	*o*	*p*	0.6	*p*	*q*	0.55	
	⋮	⋮	⋮	⋮	⋮	⋮	⋮	

Action: select next element from queue and get next tuple

Queue	δ
q	0.8

Predicate	*Answers*
q	$\langle e, 0.75\rangle$
p	$\langle e, 0.75\rangle$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
	1	a	b	1.0	m	h	0.95	
	2	c	d	0.9	m	j	0.85	
$\longrightarrow$	3	e	f	0.8	f	k	0.75	$\longleftarrow$
	4	l	m	0.7	m	n	0.65	
	5	o	p	0.6	p	q	0.55	
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Action: no new tuple for q, so expand q rule

Queue	δ
p	0.8

Predicate	*Answers*
q	$\langle e, 0.75\rangle$
p	$\langle e, 0.75\rangle$

$$q(x,s) \leftarrow p(x,s_1), s = s_1$$
$$p(x,s) \leftarrow r_1(x,y,s_1), r_2(y,z,s_2), s = \min(s_1,s_2)$$

	RecordID	r_1			r_2			
	1	a	b	1.0	m	h	0.95	
	2	c	d	0.9	m	j	0.85	
$\rightarrow$	3	e	f	0.8	f	k	0.75	$\leftarrow$
	4	l	m	0.7	m	n	0.65	
	5	o	p	0.6	p	q	0.55	
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Action: select next element from queue and get next tuple

Queue	δ
p	0.8

Predicate	*Answers*
q	$\langle e, 0.75\rangle$
p	$\langle e, 0.75\rangle$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
	1	a	b	1.0	m	h	0.95	
	2	c	d	0.9	m	j	0.85	
	3	e	f	0.8	f	k	0.75	$\leftarrow$
$\rightarrow$	4	l	m	0.7	m	n	0.65	
	5	o	p	0.6	p	q	0.55	
	⋮	⋮	⋮	⋮	⋮	⋮	⋮	

Action: get next tuple for p

Queue	δ
–	0.8

Predicate	*Answers*
q	$\langle e, 0.75\rangle$
p	$\langle e, 0.75\rangle$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
	1	a	b	1.0	m	h	0.95	
	2	c	d	0.9	m	j	0.85	
	3	e	f	0.8	f	k	0.75	$\leftarrow$
$\rightarrow$	4	l	m	0.7	m	n	0.65	
	5	o	p	0.6	p	q	0.55	
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Action: get next tuple for p

Queue	δ
—	0.8

Predicate	*Answers*
q	$\langle e, 0.75\rangle$
p	$\langle e, 0.75\rangle$, $\langle l, 0.7\rangle$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
	1	*a*	*b*	1.0	*m*	*h*	0.95	
	2	*c*	*d*	0.9	*m*	*j*	0.85	
	3	*e*	*f*	0.8	*f*	*k*	0.75	$\leftarrow$
$\rightarrow$	4	*l*	*m*	0.7	*m*	*n*	0.65	
	5	*o*	*p*	0.6	*p*	*q*	0.55	
	⋮	⋮	⋮	⋮	⋮	⋮	⋮	

Action: *p* changed, so put *q* in queue and get next tuple for it

Queue	δ
q	1

Predicate	*Answers*
q	$\langle e, 0.75\rangle$ $\langle l, 0.7\rangle$
p	$\langle e, 0.75\rangle, \langle l, 0.7\rangle$

$$q(x,s) \leftarrow p(x,s_1), s = s_1$$
$$p(x,s) \leftarrow r_1(x,y,s_1), r_2(y,z,s_2), s = \min(s_1,s_2)$$

	RecordID	r_1			r_2			
	1	a	b	1.0	m	h	0.95	
	2	c	d	0.9	m	j	0.85	
	3	e	f	0.8	f	k	0.75	$\leftarrow$
$\rightarrow$	4	l	m	0.7	m	n	0.65	
	5	o	p	0.6	p	q	0.55	
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Action: q changed, put predicates depending on q in queue

Queue	δ
–	0.8

Predicate	*Answers*
q	$\langle e, 0.75\rangle \langle l, 0.7\rangle$
p	$\langle e, 0.75\rangle, \langle l, 0.7\rangle$

$$q(x, s) \leftarrow p(x, s_1), s = s_1$$
$$p(x, s) \leftarrow r_1(x, y, s_1), r_2(y, z, s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
	1	a	b	1.0	m	h	0.95	
	2	c	d	0.9	m	j	0.85	
	3	e	f	0.8	f	k	0.75	$\leftarrow$
$\rightarrow$	4	l	m	0.7	m	n	0.65	
	5	o	p	0.6	p	q	0.55	
	⋮	⋮	⋮	⋮	⋮	⋮	⋮	

Action: queue empty, update threshold

Queue	δ
–	0.75

Predicate	*Answers*
q	$\langle e, 0.75\rangle \langle l, 0.7\rangle$
p	$\langle e, 0.75\rangle, \langle l, 0.7\rangle$

$$q(x,s) \leftarrow p(x,s_1), s = s_1$$
$$p(x,s) \leftarrow r_1(x,y,s_1), r_2(y,z,s_2), s = \min(s_1, s_2)$$

	RecordID	r_1			r_2			
	1	a	b	1.0	m	h	0.95	
	2	c	d	0.9	m	j	0.85	
	3	e	f	0.8	f	k	0.75	$\leftarrow$
$\rightarrow$	4	l	m	0.7	m	n	0.65	
	5	o	p	0.6	p	q	0.55	
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

Action: STOP, top-1 tuple score is equal or above threshold

Queue	δ
$-$	0.75

Predicate	*Answers*
q	$\langle e, 0.75\rangle \langle l, 0.7\rangle$
p	$\langle e, 0.75\rangle, \langle l, 0.7\rangle$

$$Top_1(\mathcal{P}, q) = \{\langle e, 0.75\rangle\}$$

Note: no further answer will have score above threshold δ

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